

Bash Script Tasks

1. Create a bash script to check if a directory is available or not.

```
#!/bin/bash
#to check_directory
read -p "Enter directory path: " dir

if [ -d "$dir" ]; then
    echo "Directory exists: $dir"
else
    echo "Directory does not exist: $dir"
fi
~

[root@ip-172-31-26-37 scripts]# vi check-directory.bash
[root@ip-172-31-26-37 scripts]# chmod 755 check-directory.bash
[root@ip-172-31-26-37 scripts]# ls -l check-directory.bash
-rwxr-xr-x. 1 root root 176 Aug 14 09:15 check-directory.bash
[root@ip-172-31-26-37 scripts]# ./check-directory.bash
Enter directory path: /etc
Directory exists: /etc
[root@ip-172-31-26-37 scripts]# ./check-directory.bash
Enter directory path: abc
Directory does not exist: abc
```

2. Create a bash script to create multiple files.

```
#!/bin/bash
#create multiple files.

read -p "Enter number of files to create: " n

for ((i=1; i<=n; i++))
do
    touch "file$i.txt"
done

echo "$n files created successfully."
~
~
```

```
[root@ip-172-31-26-37 scripts]# ./create-file.bash
Enter number of files to create: 5
5 files created successfully.
[root@ip-172-31-26-37 scripts]# vi create-file.bash
[root@ip-172-31-26-37 scripts]# ls -l file*.txt
-rw-r--r--. 1 root root 0 Aug 14 09:39 file1.txt
-rw-r--r--. 1 root root 0 Aug 14 09:39 file2.txt
-rw-r--r--. 1 root root 0 Aug 14 09:39 file3.txt
-rw-r--r--. 1 root root 0 Aug 14 09:39 file4.txt
-rw-r--r--. 1 root root 0 Aug 14 09:39 file5.txt
[root@ip-172-31-26-37 scripts]# |
```

3. Create a bash script to take a backup of a directory.

```
[root@ip-172-31-26-37 scripts]# ./backup-directory.bash
Enter directory path to backup: /etc
tar: Removing leading `/' from member names
Backup created successfully: /tmp/backup_20260814_095604.tar.gz
[root@ip-172-31-26-37 scripts]#
```

```
#!/bin/bash

read -p "Enter directory path to backup: " source
if [ -d "$source" ]; then
    backup="/tmp/backup_$(date +%Y%m%d_%H%M%S).tar.gz"
    tar -czf "$backup" "$source"
    echo "Backup created successfully: $backup"
else
    echo "Directory does not exist."
fi
```

4. Create a bash script to install Nginx on an EC2 server.

```
#!/bin/bash

sudo dnf install nginx -y

sudo systemctl enable nginx
sudo systemctl start nginx

echo "Nginx installed and started successfully."
sudo systemctl status nginx --no-pager
~
```

```
root@ip-172-31-26-37:~/bash-script-task/scripts
Transaction test succeeded.
Running transaction
Preparing : nginx-filesystem-1:1.30.4-1.amzn2023.0.1.noarch 1/1
Running scriptlet: nginx-filesystem-1:1.30.4-1.amzn2023.0.1.noarch 1/7
Installing : nginx-filesystem-1:1.30.4-1.amzn2023.0.1.noarch 1/7
Installing : nginx-minetypes-2.1.49-3.amzn2023.0.3.noarch 2/7
Installing : libunwind-1.4.0-5.amzn2023.0.3.x86_64 3/7
Installing : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64 4/7
Installing : nginx-core-1:1.30.4-1.amzn2023.0.1.x86_64 5/7
Installing : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 6/7
Installing : nginx-1:1.30.4-1.amzn2023.0.1.x86_64 7/7
Running scriptlet: nginx-1:1.30.4-1.amzn2023.0.1.x86_64 7/7
Verifying : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 1/7
Verifying : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64 2/7
Verifying : libunwind-1.4.0-5.amzn2023.0.3.x86_64 3/7
Verifying : nginx-1:1.30.4-1.amzn2023.0.1.x86_64 4/7
Verifying : nginx-core-1:1.30.4-1.amzn2023.0.1.x86_64 5/7
Verifying : nginx-filesystem-1:1.30.4-1.amzn2023.0.1.noarch 6/7
Verifying : nginx-minetypes-2.1.49-3.amzn2023.0.3.noarch 7/7
Installed:
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch      gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64      libunwind-1.4.0-5.amzn2023.0.3.x86_64
nginx-1:1.30.4-1.amzn2023.0.1.x86_64                  nginx-core-1:1.30.4-1.amzn2023.0.1.x86_64      nginx-filesystem-1:1.30.4-1.amzn2023.0.1.noarch
nginx-minetypes-2.1.49-3.amzn2023.0.3.noarch
Complete!
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service - /usr/lib/systemd/system/nginx.service.
Nginx installed and started successfully.
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
   Active: active (running) since Fri 2026-08-14 10:05:43 UTC; 79ms ago
     Process: 6712 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
     Process: 6743 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
     Process: 6783 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
   Main PID: 6827 (nginx)
    Tasks: 3 (limit: 1059)
   Memory: 3.4M
      CPU: 75ms
   CGroup: /system.slice/nginx.service
           └─6827 "nginx: master process /usr/sbin/nginx"
             └─6828 "nginx: worker process"
               └─6830 "nginx: worker process"

Aug 14 10:05:43 ip-172-31-26-37.ec2.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Aug 14 10:05:43 ip-172-31-26-37.ec2.internal nginx[6743]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Aug 14 10:05:43 ip-172-31-26-37.ec2.internal nginx[6743]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Aug 14 10:05:43 ip-172-31-26-37.ec2.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
[root@ip-172-31-26-37 scripts]#
```

5. Create a bash script to install Apache Tomcat on an EC2 server.

```
root@ip-172-31-26-37:~/bash-script-task/scripts
#!/bin/bash

sudo dnf install java-21-amazon-corretto -y

cd /opt

sudo wget https://d1cdn.apache.org/tomcat/tomcat-10/v10.1.52/bin/apache-tomcat-10.1.52.tar.gz

sudo tar -xzf apache-tomcat-10.1.52.tar.gz

sudo mv apache-tomcat-10.1.52 tomcat

sudo chmod +x /opt/tomcat/bin/*.sh

sudo /opt/tomcat/bin/startup.sh

echo "Apache Tomcat installed and started successfully."
~
~

root@ip-172-31-26-37:~/bash-script-task/scripts
Verifying : libXrender-0.9.11-6.amzn2023.0.1.x86_64 27/35
Verifying : libXt-1.3.0-3.amzn2023.0.1.x86_64 28/35
Verifying : libXtst-1.2.5-1.amzn2023.0.1.x86_64 29/35
Verifying : libbrotli-1.0.9-4.amzn2023.0.2.x86_64 30/35
Verifying : libjpeg-turbo-2.1.4-2.amzn2023.0.5.x86_64 31/35
Verifying : libpng-2.1.6-37-10.amzn2023.0.13.x86_64 32/35
Verifying : libxcb-1.17.0-1.amzn2023.0.1.x86_64 33/35
Verifying : pixman-0.43.4-1.amzn2023.0.4.x86_64 34/35
Verifying : xml-common-0.6.3-56.amzn2023.0.2.noarch 35/35

Installed:
alsa-lib-1.2.7.2-1.amzn2023.0.3.x86_64
dejavu-sans-fonts-2.37-16.amzn2023.0.2.noarch
dejavu-serif-fonts-2.37-16.amzn2023.0.2.noarch
fonts-filesystem-1.2.0-5-12.amzn2023.0.2.noarch
giflib-5.2.1-9.amzn2023.0.4.x86_64
google-noto-sans-vf-fonts-20240401-1.amzn2023.0.2.noarch
harfbuzz-7.0.0-2.amzn2023.0.2.x86_64
java-21-amazon-corretto-headless-1.21.0.12-8-1.amzn2023.1.x86_64
langpacks-core-font-en-3.0-21.amzn2023.0.4.noarch
libSM-1.2.4-3.amzn2023.0.1.x86_64
libX11-common-1.8.10-2.amzn2023.0.1.noarch
libXext-1.3.6-1.amzn2023.0.1.x86_64
libXinerama-1.1.5-6.amzn2023.0.1.x86_64
libXrender-0.9.11-6.amzn2023.0.1.x86_64
libXtst-1.2.5-1.amzn2023.0.1.x86_64
libjpeg-turbo-2.1.4-2.amzn2023.0.5.x86_64
libxcb-1.17.0-1.amzn2023.0.1.x86_64
xml-common-0.6.3-56.amzn2023.0.2.noarch
cairo-1.18.0-4.amzn2023.0.3.x86_64
dejavu-sans-mono-fonts-2.37-16.amzn2023.0.2.noarch
fontconfig-2.13.94-2.amzn2023.0.2.x86_64
freetype-2.13.2-5.amzn2023.0.2.x86_64
google-noto-fonts-common-20240401-1.amzn2023.0.2.noarch
graphite2-1.3.14-7.amzn2023.0.3.x86_64
java-21-amazon-corretto-1:21.0.12-8-1.amzn2023.1.x86_64
javapackages-filesystem-6.0.0-7.amzn2023.0.6.noarch
libICE-1.1.1-3.amzn2023.0.1.x86_64
libX11-1.8.10-2.amzn2023.0.1.x86_64
libXau-1.0.11-6.amzn2023.0.1.x86_64
libX1-1.8.2-1.amzn2023.0.1.x86_64
libXrandr-1.5.4-3.amzn2023.0.1.x86_64
libXt-1.3.0-3.amzn2023.0.1.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64
libpng-2.1.6-37-10.amzn2023.0.13.x86_64
pixman-0.43.4-1.amzn2023.0.4.x86_64

Complete!
--2026-08-14 10:12:40-- https://d1cdn.apache.org/tomcat/tomcat-10/v10.1.52/bin/apache-tomcat-10.1.52.tar.gz
Resolving d1cdn.apache.org (d1cdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to d1cdn.apache.org (d1cdn.apache.org)[151.101.2.132]:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-08-14 10:12:40 ERROR 404: Not Found.

tar (child): apache-tomcat-10.1.52.tar.gz: Cannot open: No such file or directory
tar (child): Error is not recoverable: exiting now
tar: Child returned status 2
tar: Error is not recoverable: exiting now
mv: cannot stat 'apache-tomcat-10.1.52': No such file or directory
chmod: cannot access '/opt/tomcat/bin/*.sh': No such file or directory
sudo: /opt/tomcat/bin/startup.sh: command not found
Apache Tomcat installed and started successfully.
[root@ip-172-31-26-37 scripts]# ps -ef | grep tomcat
root      28844      2222  0 10:13 pts/3    00:00:00 grep --color=auto tomcat
[root@ip-172-31-26-37 scripts]#
```

6. Create a bash script to check if the Nginx service is running, if not running then script should start the service.

```
[root@ip-172-31-26-37 scripts]# vi check-nginx.bash
[root@ip-172-31-26-37 scripts]# chmod 755 check-nginx.bash
[root@ip-172-31-26-37 scripts]# ./check-nginx.bash
Nginx is running.
[root@ip-172-31-26-37 scripts]#

root@ip-172-31-26-37:~/bash-script-task/scripts
#!/bin/bash

if systemctl is-active --quiet nginx
then
    echo "Nginx is running."
else
    echo "Nginx is not running. Starting Nginx..."
    sudo systemctl start nginx
    echo "Nginx started successfully."
fi
```

7. Create a bash script for a calculator.

```
[root@ip-172-31-26-37 scripts]# vi calculator-bash
[root@ip-172-31-26-37 scripts]# chmod 755 calculator-bash
[root@ip-172-31-26-37 scripts]# ./calculator-bash
Enter first number: 5
Enter operator (+ - * /): +
Enter second number: 25
Result: 30
[root@ip-172-31-26-37 scripts]# |
```

root@ip-172-31-26-37:~/bash-script-task/scripts

```
#!/bin/bash

read -p "Enter first number: " a
read -p "Enter operator (+ - * /): " op
read -p "Enter second number: " b

case $op in
    +)
        echo "Result: $((a + b))"
        ;;
    -)
        echo "Result: $((a - b))"
        ;;
    \*)
        echo "Result: $((a * b))"
        ;;
    /)
        if [ "$b" -eq 0 ]; then
            echo "Cannot divide by zero."
        else
            echo "Result: $((a / b))"
        fi
        ;;
    *)
        echo "Invalid operator."
        ;;
esac

~
```

8. Create a bash script to **check if a directory exists**, if not then create a directory.

```
[root@ip-172-31-26-37 scripts]# vi create-directory,.bash
[root@ip-172-31-26-37 scripts]# chmod 755 create-directory,.bash
[root@ip-172-31-26-37 scripts]# ./create-directory,.bash
Enter directory name: mkdir
Directory created successfully.
[root@ip-172-31-26-37 scripts]# |
```

root@ip-172-31-26-37:~/bash-script-task/scripts

```
#!/bin/bash

read -p "Enter directory name: " dir

if [ -d "$dir" ]; then
    echo "Directory already exists."
else
    mkdir -p "$dir"
    echo "Directory created successfully."
fi

~
~
```

9. Create a bash script to **delete the last 3 lines of a file**.

```
[root@ip-172-31-26-37 scripts]# vi delete-3-lines.bash
[root@ip-172-31-26-37 scripts]# chmod 755 delete-3-lines.bash
[root@ip-172-31-26-37 scripts]# cd ~/bash-script-task/scripts
[root@ip-172-31-26-37 scripts]# printf "line 1\nline 2\nline 3\nline 4\nline 5\nline 6\n" >sample.txt
[root@ip-172-31-26-37 scripts]# cat sample.txt
line 1
line 2
line 3
line 4
line 5
line 6
[root@ip-172-31-26-37 scripts]# ./delete-3-lines.bash
Enter file name: sample.txt
Last 3 lines deleted successfully.
[root@ip-172-31-26-37 scripts]# cat sample.txt
line 1
line 2
line 3
[root@ip-172-31-26-37 scripts]# |
```

```
root@ip-172-31-26-37:~/bash-script-task/scripts
#!/bin/bash

read -p "Enter file name: " file

if [ -f "$file" ]; then
    head -n -3 "$file" > temp.txt
    mv temp.txt "$file"
    echo "Last 3 lines deleted successfully."
else
    echo "File does not exist."
fi
~
~
```

10. Bash script to monitor cpu and if it is more than 80% then send email notification.

```
[root@ip-172-31-26-37 scripts]# vi cpu-monitor.bash
[root@ip-172-31-26-37 scripts]# chmod 755 cpu-monitor.bash
[root@ip-172-31-26-37 scripts]# ./cpu-monitor.bash
Current CPU usage: 3.2%
CPU usage is normal.
[root@ip-172-31-26-37 scripts]# |
```

```
root@ip-172-31-26-37:~/bash-script-task/scripts
#!/bin/bash

CPU=$(top -bn1 | awk '/Cpu\s\)/ {print 100 - $8}')
CPU_INT=${CPU%.*}

echo "Current CPU usage: ${CPU}%"

if [ "$CPU_INT" -gt 80 ]; then
    echo "CPU usage is ${CPU}% on $(hostname)" | mail -s "CPU Usage Alert" your@email.com
    echo "Email notification sent."
else
    echo "CPU usage is normal."
fi
~
~
```

11. Bash script to monitor disk space and if it is more than 80% then send email notification.

```
[root@ip-172-31-26-37 scripts]# vi disk-monitor.bash
[root@ip-172-31-26-37 scripts]# chmod 755 disk-monitor.bash
[root@ip-172-31-26-37 scripts]# ./disk-monitor.bash
Current disk usage: 25%
Disk usage is normal.
[root@ip-172-31-26-37 scripts]# |
```

```

root@ip-172-31-26-37:~/bash-script-task/scripts
#!/bin/bash

DISK=$(df / | awk 'NR==2 {print $5}' | sed 's/%//')

echo "Current disk usage: ${DISK}%"

if [ "$DISK" -gt 80 ]; then
    echo "Disk usage is ${DISK}% on $(hostname)" | mail -s "Disk Space Alert" your@email.com
    echo "Email notification sent."
else
    echo "Disk usage is normal."
fi

```

12. Bash script to monitor memory and if it is more than 80% then send email notification.

```

root@ip-172-31-26-37:~/bash-script-task/scripts
#!/bin/bash

MEM=$(free | awk '/Mem:/ {printf "%.0f", ($3/$2)*100}')

echo "Current memory usage: ${MEM}%"

if [ "$MEM" -gt 80 ]; then
    echo "Memory usage is ${MEM}% on $(hostname)" | mail -s "Memory Usage Alert" your@email.com
    echo "Email notification sent."
else
    echo "Memory usage is normal."
fi

~
[root@ip-172-31-26-37 scripts]# vi memory-monitor.bash
[root@ip-172-31-26-37 scripts]# chmod 755 memory-monitor.bash
[root@ip-172-31-26-37 scripts]# ./memory-monitor.bash
Current memory usage: 22%
Memory usage is normal.
[root@ip-172-31-26-37 scripts]# |

```

Exercise: Crontab Entries

1. April 5th Midnight
2. 5th of Every November, January, June if it is a Thursday
3. At 05 and 27th minutes of 9, 10, 11 hours every day
4. 34 minutes of 9th hour on 15th August
5. Every midnight
6. Every Weekend (Saturday night 11:59 PM)
7. After every reboot

```

[root@ip-172-31-26-37 crontab]# cat -n cron_entries.txt
 1 0 0 5 4 * echo "April 5 midnight" >> /tmp/cron-test.log
 2 0 0 5 1,6,11 4 echo "5th Jan Jun Nov Thursday" >> /tmp/cron-test.log
 3 5,27 9-11 * * * echo "9-11 hours 05 and 27 minutes" >> /tmp/cron-test.log
 4 34 9 15 8 * echo "August 15 9:34 AM" >> /tmp/cron-test.log
 5 0 0 * * * echo "Every midnight" >> /tmp/cron-test.log
 6 59 23 * * 6 echo "Saturday 11:59 PM" >> /tmp/cron-test.log
 7 @reboot echo "System rebooted" >> /tmp/reboot-test.log
[root@ip-172-31-26-37 crontab]# crontab -T cron_entries.txt
No syntax issues were found in the crontab file.
[root@ip-172-31-26-37 crontab]# crontab cron_entries.txt
[root@ip-172-31-26-37 crontab]# crontab -l
0 0 5 4 * echo "April 5 midnight" >> /tmp/cron-test.log
0 0 5 1,6,11 4 echo "5th Jan Jun Nov Thursday" >> /tmp/cron-test.log
5,27 9-11 * * * echo "9-11 hours 05 and 27 minutes" >> /tmp/cron-test.log
34 9 15 8 * echo "August 15 9:34 AM" >> /tmp/cron-test.log
0 0 * * * echo "Every midnight" >> /tmp/cron-test.log
59 23 * * 6 echo "Saturday 11:59 PM" >> /tmp/cron-test.log
@reboot echo "System rebooted" >> /tmp/reboot-test.log
[root@ip-172-31-26-37 crontab]# |

```

Push the tasks to a GitHub repository(explre this for now).

GitHub Repository : <https://github.com/chvkarthik-87/bash-script-task>

Project Documentation : <https://github.com/chvkarthik-87/bash-script-task/commit/8c247fc77b045fbd40717bf9ea05fa88cc0063fe>